

Jivan Singh

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EDUCATION

Illinois Institute of technology Chicago, IL, United States	<i>Expected May 2026</i>
<i>Master of Science in Computer Science</i>	<i>GPA: 3.40/4.00</i>
University of Mumbai Mumbai, India	<i>May 2023</i>
<i>Bachelor of Engineering in Computer Engineering</i>	<i>GPA: 3.80/4.00</i>

PROFESSIONAL EXPERIENCE

Software Developer Crayon Infotech Pvt Ltd., India	<i>Aug 2023 – Jul 2024</i>
<i>Tech Stack: PHP CodeIgniter Laravel MVC Python Git SQL/NoSQL PostgreSQL RESTful API Flask API</i>	
- Shipped 5+ full-stack apps (Laravel/CodeIgniter) serving 2K+ active users across multiple environments; sustained 99.9% uptime by refining DB indices, tightening query plans, and closing 100% of assigned JIRA tickets on schedule - Developed and deployed REST/Flask APIs that enabled integrations with 3 external vendors and internal systems, reducing integration-related support tickets by ~20% and cutting onboarding time from 2 weeks → 3 days - Collaborated with teams to deploy 5+ applications, refined the codebase to ensure stability and reduced bugs by 20% - Designed and optimized SQL queries to boost performance and ensure efficient data retrieval from large-scale databases - Engineered and maintained backend systems using Python, Flask, and PostgreSQL, optimized data flow, improved query performance by 25%, and ensured high availability and scalability for real-time web applications	
Software Developer Intern Lab Systems Pvt. Ltd., India	<i>Jul 2022 – Feb 2023</i>
<i>Tech Stack: Python Blockchain Architecture Blockchain Nodes Hashing Solidity Smart Contracts Ganache IPFS</i>	
- Programmed Python scripts and simplified OSINT workflows to analyze transactions and support forensic research - Handled more than 20 TB of blockchain data using indexing and engineering methods, ensuring high-throughput analytics - Provisioned and maintained 3+ cryptocurrency nodes, enabling secure on-premises investigations in stealth mode - Mentored a team of 12 professionals in blockchain investigation methodologies to strengthen technical capabilities	

PROJECTS

Synthetic Monitoring Platform Tech Stack: Python, Prometheus, Grafana, PromQL, Alertmanager, YAML
- Formulated a Python exporter to capture server uptime, latency, and packet loss, with metrics stored in Prometheus - Configured Prometheus to evaluate custom alerting rules and manage time-series monitoring data effectively - Crafted Grafana dashboards powered by PromQL to visualize real-time performance and system health - Implemented alerting rules and integrated Alertmanager with Gmail SMTP for automated incident notifications
Time-Series Forecasting Tech Stack: TensorFlow/Keras, Facebook Prophet, StatsModels, Flask, React, Docker, GCP
- Built a pipeline to ingest GitHub data (issues, PRs, commits, branches, contributors, releases) from 2 years - Engineered Flask microservices for model training and forecasting with a React dashboard, containerized using Docker - Launched services on Google Cloud Run with CI/CD pipelines, improving efficiency, scalability, and reliability by 40% - Delivered insights into repository health, enabling prediction of contributor trends and proactive project management
AI-Powered Collaborative Code Editor Tech Stack: React.js, Node.js, Express.js, Socket.IO, CodeMirror, OpenAI API
- Designed and implemented a real-time collaborative editor using Socket.IO, enabling multi-user sessions with file tracking - Built a browser-based compiler for Python, C, C++, and Java with AI summarization to simplify code understanding - Integrated an OpenAI-powered chatbot for real-time debugging and code queries, improving error resolution by 60% - Developed a React.js frontend with dedicated views for editor, chat, compiler, and session control to enhance usability

SKILLS & CERTIFICATIONS

- Programming Languages:** Python, C++/C, Java, JavaScript, TypeScript, SQL, NoSQL, PHP, HTML, CSS, SASS
- Frameworks & Libraries:** Next.js, React.js, Node.js, Express.js, CodeIgniter, Laravel, Flask, Django, Pandas, Matplotlib
- Tools & Platforms:** Git/GitHub, Docker, CI/CD (GitHub Actions), AWS (EC2, S3, Lambda), Firebase, Postman, REST APIs, JIRA, Slack, Vercel
- Systems & Concepts:** Data Structures & Algorithms, Object-Oriented Programming, System Design, Operating Systems, Networking, Agile Methodologies, Software Project Management, API Integration
- Certifications:** AI Agents in LangGraph (DeepLearning.AI), AI Agentic Design Patterns with AutoGen (DeepLearning.AI), AWS Academy Cloud Foundations, Advanced Software Engineering (Walmart USA)

PATENT & PUBLICATIONS

- Medilog** – (Granted), German Patent File No - 20 2023 102 823.3 | Granted: September 19, 2023
Designed and developed a full-stack web application for tracking prescription drugs using Blockchain, enabling secure and transparent pharmaceutical supply chain management. Improved traceability by 85%, reduced counterfeit drug incidents by 70% in pilot deployments, and streamlined compliance reporting by automating end-to-end regulatory processes [Patent Link](#)